TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2025-26(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester
Semester: V
Programme: B. Tech Information Technology
Course: Theory of Computation

Time: 3 Hour　　**Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33502	Course Outcome
CO1	To illustrate finite state machines to solve problems in computing.
CO2	To classify theoretical foundations of computer science from the perspective of formal languages.
CO3	To familiarize Regular grammars and Context Free Grammars.
CO4	To enhance students' ability to understand and conduct mathematical proofs for computation and algorithms.
CO5	To explain hierarchy of problems arising in computer science.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define DFA.	CO1	2	2
b.	Define RE.	CO2	2	2
c.	Define CFG.	CO3	2	2
d.	Define TM.	CO4	3	2
e.	Define undecidability.	CO5	2	2
Group B				

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain DFA and NFA.	CO1	2	5
b.	Convert NFA to DFA.	CO1	2	5
c.	Explain FA minimization.	C01	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain pumping lemma.	CO2	2	5
b.	Convert RE to FA.	CO2	3	5
c.	Explain closure properties.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain CFG and PDA.	CO3	2	5
b.	Convert CFG to GNF with taking some example also.	CO3	3	5
c.	Explain ambiguity.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain TM and variants.	CO4	2	5
b.	Explain recursive vs RE languages.	CO4	3	5
c.	Explain LBA.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Halting problem.	CO5	2	5
b.	Explain Cook's theorem.	CO5	3	5
c.	Explain recursive functions.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create